

API INTEGRATION PROJECT COSMOEXPLORER DOCUMENTATION

Internship: Frontend Developer Intern at ALFIDO TECH

Task: Task 3 – API Integration Project – CosmoExplorer

Developed By: Kullachi Akshaya

Technology Used: HTML5, CSS3, JavaScript

Hosting Platform: <https://akshaya79896-debug.github.io/ALFIDO-TECH-TASK3/>

Source Code Repository:

<https://github.com/akshaya79896-debug/ALFIDO-TECH-TASK3.git>

Project Title

CosmoExplorer – Space Data Visualization Platform

A responsive and interactive web application that integrates multiple NASA REST APIs to display real-time space data, images, and scientific information.

Objective

The objective of this project is to demonstrate API integration using JavaScript Fetch API by retrieving live data from NASA Open APIs and displaying it dynamically on a web page. The application includes loading indicators, error handling, search functionality, favourites management, and theme switching.

Description

CosmoExplorer is a futuristic space exploration website that allows users to discover NASA's latest astronomy content and scientific data.

The application provides:

-  Astronomy Picture of the Day (APOD)
-  Search NASA's image database
-  Featured Space Gallery
-  Near-Earth Asteroid Tracker
-  Save favourite images using Local Storage
-  Dark/Light Theme Toggle

The project combines multiple APIs and presents scientific information in a visually attractive and user-friendly interface.

Features

- Dynamic API integration using Fetch API
- Loading and error states
- Search functionality
- Featured image gallery
- Real-time asteroid information
- Favourites system using Local Storage
- Dark and light theme toggle
- Fully responsive design
- Modern glass morphism interface

APIs Used

1. Astronomy Picture of the Day (APOD) API

Provides NASA's daily astronomy image or video along with a scientific explanation.

2. NASA Image and Video Library API

Used to search space-related images and generate the Featured Space Gallery.

3. Near Earth Object (NEO) API

Displays information about asteroids currently approaching Earth.

Short Write-Up on API Used

This project uses NASA Open APIs, which are publicly available REST APIs that provide real-time scientific data and images from NASA missions. The APIs return data in JSON format, which is fetched using JavaScript's Fetch API and dynamically rendered in the browser.

The main APIs used are:

- APOD API for daily astronomy images
- NASA Image Search API for searching space photographs
- NEO API for tracking near-Earth asteroids

These APIs make the application dynamic, educational, and visually engaging.

Technologies Used

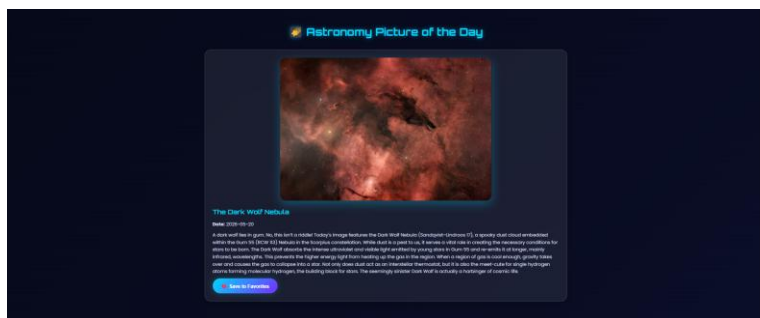
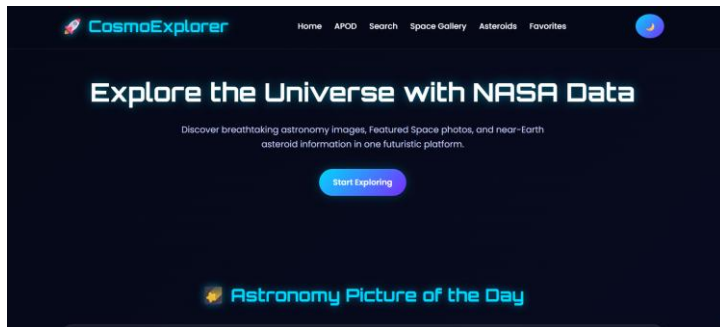
Technology	Purpose
HTML5	Website structure
CSS3	Styling, animations, responsive design
JavaScript (ES6)	API integration and interactivity
Fetch API	HTTP requests
Local Storage	Save favorite images
NASA Open APIs	Live space data
Git & GitHub	Version control
GitHub Pages	Deployment

Project Structure

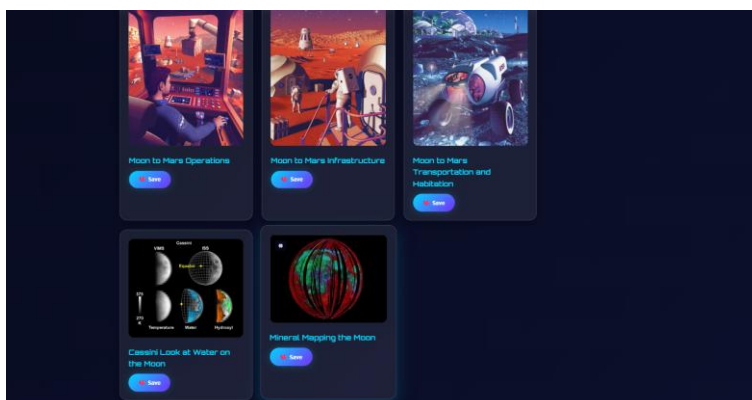
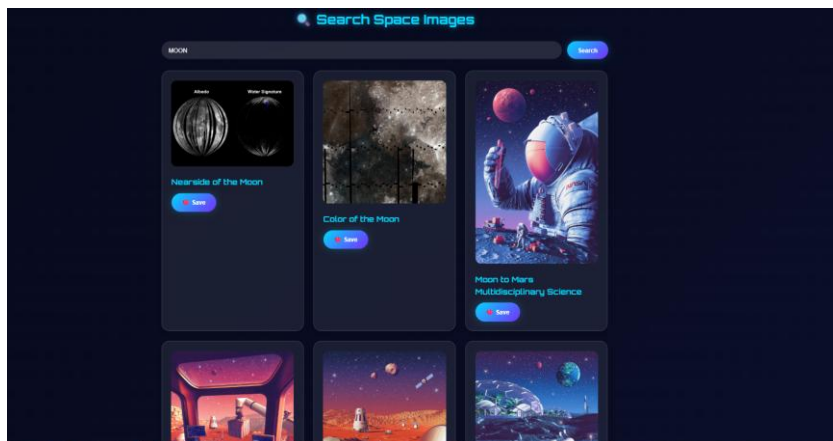
CosmoExplorer/
| — index.html
| — style.css
| — script.js
| — README.md

SCREENSHOTS

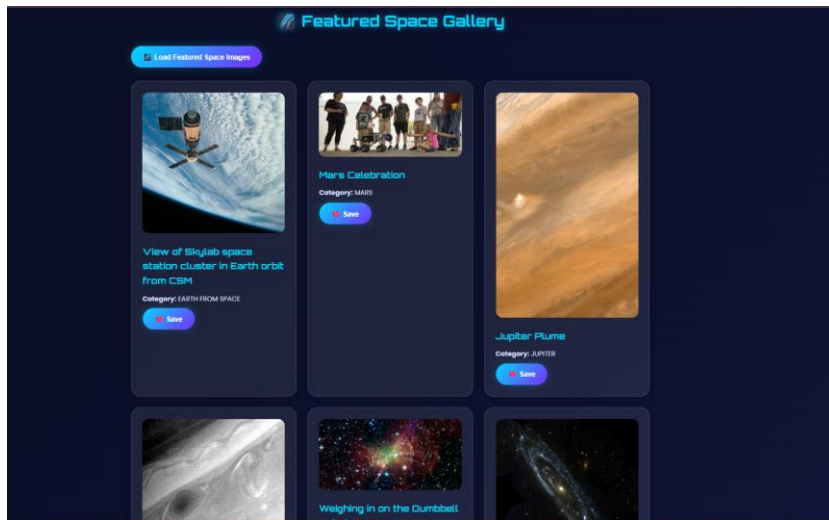
1.Home page with Astronomy Picture of the Day



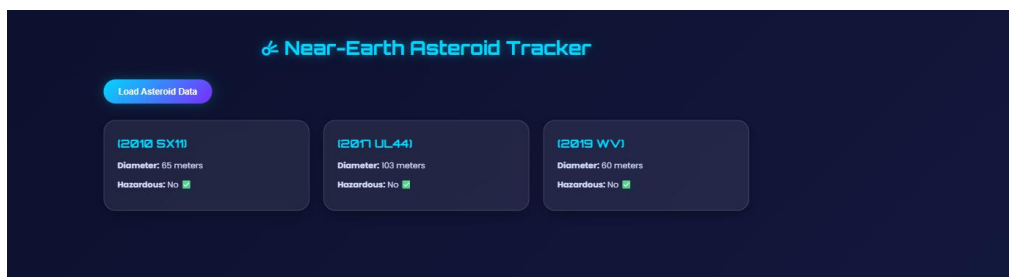
2. NASA Image Search results



3. Featured Space Gallery



4. Near-Earth Asteroid Tracker



5. Favourites Section



Code Snippets

Fetch API Example:

```
const response = await fetch(  
  `https://api.nasa.gov/planetary/apod?api_key=${API_KEY}`  
);  
const data = await response.json();
```

Saving Favourites:

```
localStorage.setItem("favorites", JSON.stringify(favorites));
```

Theme Toggle:

```
document.body.classList.toggle("light-mode");
```

Commands Used

```
git init  
git add .  
git commit -m "Initial commit"  
git branch -M main  
git remote add origin  
https://github.com/akshaya79896-debug/ALFIDO-TECH-TASK3
```

```
git push -u origin main
```

GitHub Repository Link

<https://github.com/akshaya79896-debug/ALFIDO-TECH-TASK3>

Live Hosted Website Link

<https://akshaya79896-debug.github.io/ALFIDO-TECH-TASK3/>

Deployment Process

Step 1: Create Repository

Create a new repository on GitHub.

Step 2: Upload Project Files

Upload:

- index.html
- style.css
- script.js

Step 3: Enable GitHub Pages

Go to **Settings** → **Pages**, select the main branch and /root folder, then save.

Step 4: Access Live Website

GitHub Pages generates a public URL for the project.

Testing

The application was tested for:

- API response handling
- Loading indicators
- Error messages
- Search functionality
- Favourites storage
- Theme toggle
- Responsive design
- Cross-browser compatibility

Browsers tested:

- [Google Chrome](#)
- [Mozilla Firefox](#)
- [Microsoft Edge](#)

Learning Outcomes

Through this project, I learned:

- Working with REST APIs
- Using Fetch API and async/await
- Handling JSON data
- Implementing loading and error states
- Using Local Storage
- Creating responsive user interfaces
- Deploying projects using GitHub Pages

Conclusion

CosmoExplorer successfully demonstrates the integration of multiple NASA APIs to create a dynamic and educational web application. The project combines scientific content, responsive design, and interactive features to provide an engaging user experience while strengthening my frontend development and API integration skills.

Declaration

I hereby declare that this project was developed by me as part of my Frontend Developer Internship at Alfido Tech.